

Actionability Affordance Factorization

Follow-up on actionability_algebra_v4: the fused 4-way permission label is weak, but the primitive affordance bits are materially cleaner.

DATE	PHASE	PROMPTS	TEST GROUPS
21 March 2026	actionability_algebra_v4	96 captured	6 held-out groups

MAIN READ

permission_mode is probably the wrong probe target. The model appears to represent primitive affordance bits more cleanly than the fused 4-way mode.

V4 PERMISSION MODE	V4 CAN BUY	V4 CAN SELL	TOP-SYMBOL INVARIANCE
0.458	0.792	0.750	1.000
bal acc	portfolio_eos L18	constraints_eos L41	unchanged

Why This Matters

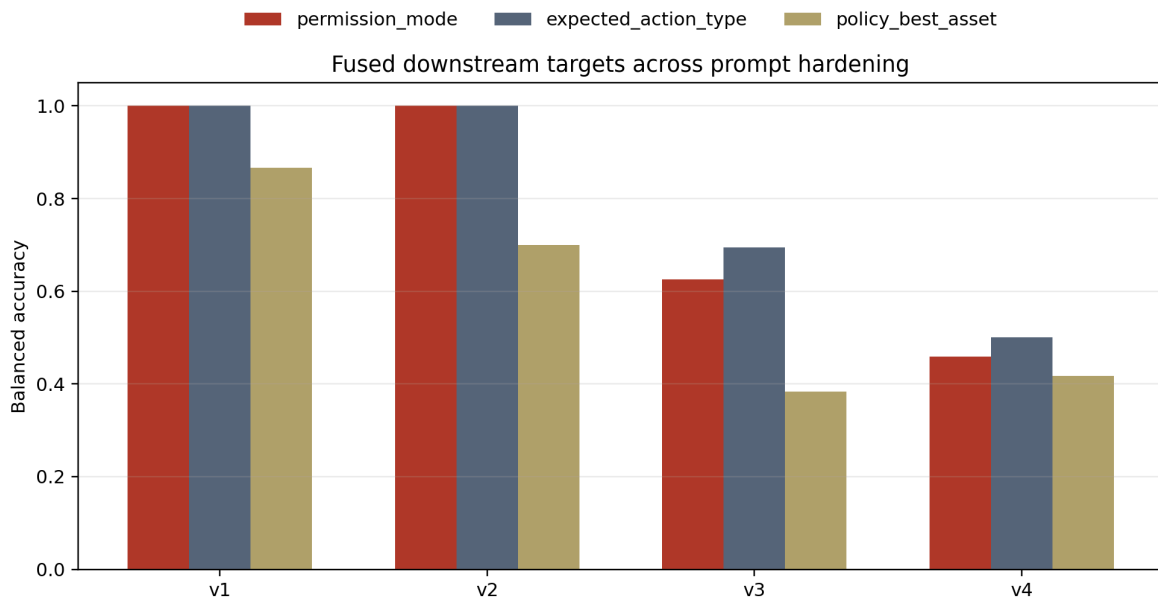
The counting-manifolds paper explicitly warns that early probe attempts can fail when the target is a fused variable rather than the quantity the model actually represents cleanly.

That seems to be happening here.

- The fused 4-way permission_mode label is weak under hardened prompt wording.
- But the primitive bits can_buy, can_sell, and observe_vs_act survive materially better on the same pooled activations.

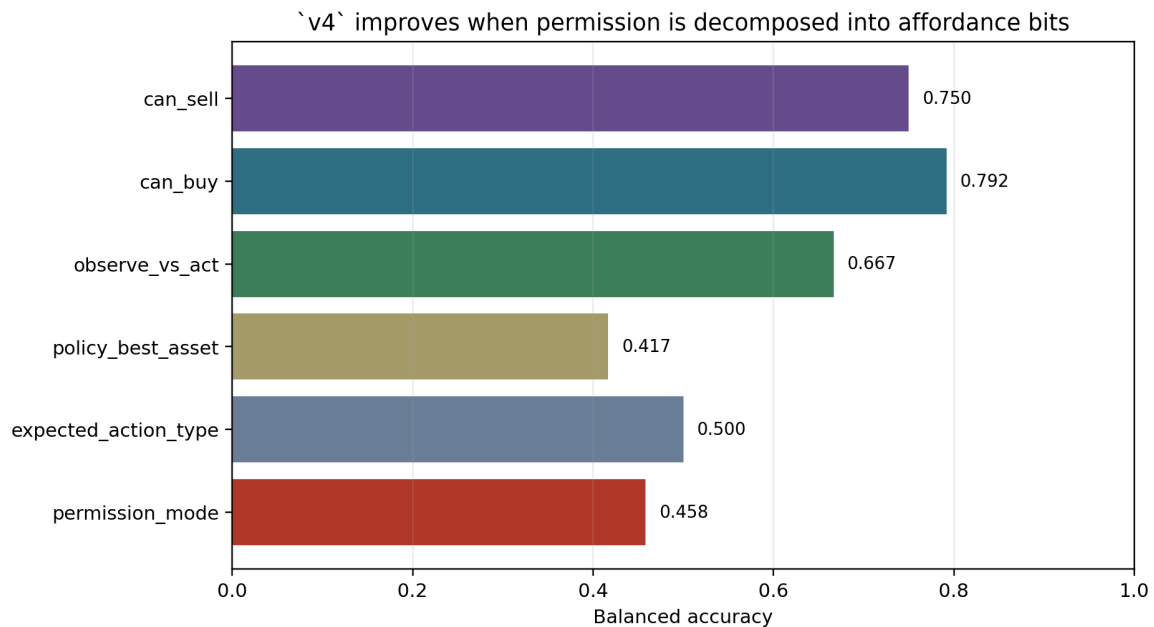
This is the first result in the actionability line that suggests a more precise mechanistic target rather than just a weaker or stronger version of the same story.

Phase Progression



Fused downstream targets across v1 through v4. v2 looked strong, but prompt hardening in v3 and v4 degraded the fused permission-side reads sharply.

v4 Changes The Story

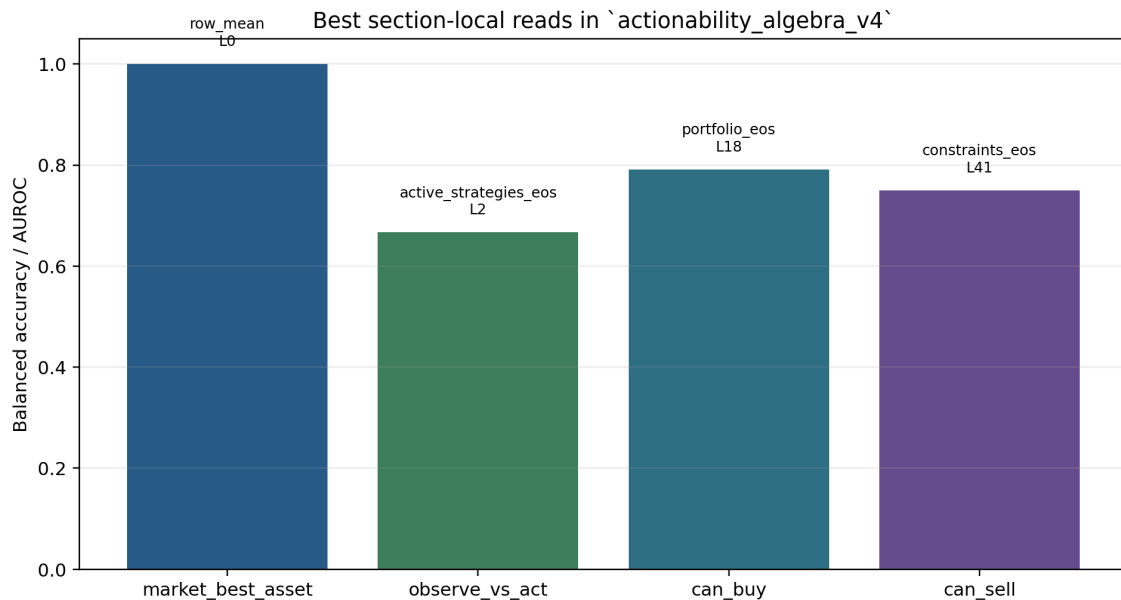


On v4, decomposed affordance bits outperform the fused 4-way mode by a large margin.

Target	Best State	Balanced Accuracy
permission_mode	portfolio_eos L13	0.458

Target	Best State	Balanced Accuracy
expected_action_type	last_token L13	0.500
observe_vs_act	active_strategies_eos L2	0.667
can_buy	portfolio_eos L18	0.792
can_sell	constraints_eos L41	0.750

Section Read



The primitive affordance bits land in plausible different downstream sections. Buyability is strongest in portfolio state, sellability in constraints, and act-vs-observe near the strategies section.

Interpretation

SUPPORTED NOW

The actionability computation looks more factorized than the earlier fused-label framing suggested.

NOT SUPPORTED YET

A clean monolithic permission_mode representation is still not established.

Best current interpretation:

- early market preference is still robust and easy
- downstream permission is not one crisp fused state

- instead, the model appears to carry primitive affordances separately
- that means the next probe family should target `can_buy`, `can_sell`, and `observe_vs_act`, not just `permission_mode`

Next Step

The best next experiment is no longer another fused permission rerun. It is:

- carry the factorized labels into the next synthetic prompt-hardening pass
- test whether `can_buy` and `can_sell` survive additional paraphrase and row-format variation
- then bridge those binary affordance probes into real DX paired reruns

Actionability Affordance Factorization — 21 March 2026. Synthetic v4 plus decomposed affordance probes on the same pooled residuals.